# 1 Preliminaries to Complex Analysis

> The sweeping development of mathematics during the last two centuries is due in large part to the introduction of complex numbers; paradoxically, this is based on the seemingly absurd notion that there are numbers whose squares are negative.
>
> *E. Borel,* 1952

This chapter is devoted to the exposition of basic preliminary material which we use extensively throughout of this book.

We begin with a quick review of the algebraic and analytic properties of complex numbers followed by some topological notions of sets in the complex plane. (See also the exercises at the end of Chapter 1 in Book I.)

Then, we define precisely the key notion of holomorphicity, which is the complex analytic version of differentiability. This allows us to discuss the Cauchy-Riemann equations, and power series.

Finally, we define the notion of a curve and the integral of a function along it. In particular, we shall prove an important result, which we state loosely as follows: if a function $f$ has a primitive, in the sense that there exists a function $F$ that is holomorphic and whose derivative is precisely $f$, then for any closed curve $\gamma$

$$\int_\gamma f(z)\,dz = 0.$$

This is the first step towards Cauchy's theorem, which plays a central role in complex function theory.

## 1 Complex numbers and the complex plane

Many of the facts covered in this section were already used in Book I.

### 1.1 Basic properties

A complex number takes the form $z = x + iy$ where $x$ and $y$ are real, and $i$ is an imaginary number that satisfies $i^2 = -1$. We call $x$ and $y$ the

**real part** and the **imaginary part** of $z$, respectively, and we write

$$x = \mathrm{Re}(z) \quad \text{and} \quad y = \mathrm{Im}(z).$$

The real numbers are precisely those complex numbers with zero imaginary parts. A complex number with zero real part is said to be **purely imaginary**.

Throughout our presentation, the set of all complex numbers is denoted by $\mathbb{C}$. The complex numbers can be visualized as the usual Euclidean plane by the following simple identification: the complex number $z = x + iy \in \mathbb{C}$ is identified with the point $(x, y) \in \mathbb{R}^2$. For example, 0 corresponds to the origin and $i$ corresponds to $(0, 1)$. Naturally, the $x$ and $y$ axis of $\mathbb{R}^2$ are called the **real axis** and **imaginary axis**, because they correspond to the real and purely imaginary numbers, respectively. (See Figure 1.)

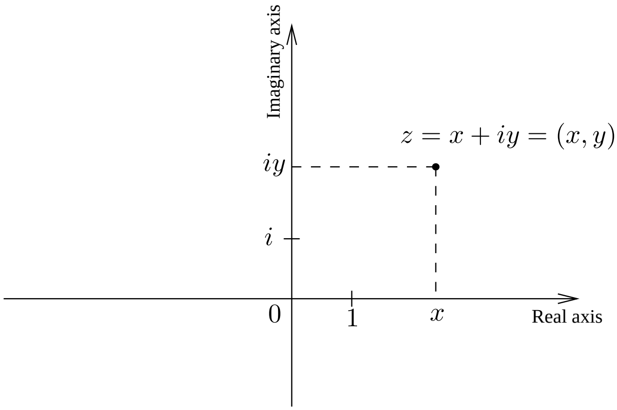


**Figure 1.** The complex plane

The natural rules for adding and multiplying complex numbers can be obtained simply by treating all numbers as if they were real, and keeping in mind that $i^2 = -1$. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2),$$

and also

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2 \\ &= (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2). \end{aligned}$$

If we take the two expressions above as the definitions of addition and multiplication, it is a simple matter to verify the following desirable properties:

- Commutativity: $z_1 + z_2 = z_2 + z_1$ and $z_1 z_2 = z_2 z_1$ for all $z_1, z_2 \in \mathbb{C}$.
- Associativity: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$; and $(z_1 z_2) z_3 = z_1(z_2 z_3)$ for $z_1, z_2, z_3 \in \mathbb{C}$.
- Distributivity: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ for all $z_1, z_2, z_3 \in \mathbb{C}$.

Of course, addition of complex numbers corresponds to addition of the corresponding vectors in the plane $\mathbb{R}^2$. Multiplication, however, consists of a rotation composed with a dilation, a fact that will become transparent once we have introduced the polar form of a complex number. At present we observe that multiplication by $i$ corresponds to a rotation by an angle of $\pi/2$.

The notion of length, or absolute value of a complex number is identical to the notion of Euclidean length in $\mathbb{R}^2$. More precisely, we define the **absolute value** of a complex number $z = x + iy$ by

$$|z| = (x^2 + y^2)^{1/2},$$

so that $|z|$ is precisely the distance from the origin to the point $(x, y)$. In particular, the triangle inequality holds:

$$|z + w| \leq |z| + |w| \qquad \text{for all } z, w \in \mathbb{C}.$$

We record here other useful inequalities. For all $z \in \mathbb{C}$ we have both $|\mathrm{Re}(z)| \leq |z|$ and $|\mathrm{Im}(z)| \leq |z|$, and for all $z, w \in \mathbb{C}$

$$||z| - |w|| \leq |z - w|.$$

This follows from the triangle inequality since

$$|z| \leq |z - w| + |w| \quad \text{and} \quad |w| \leq |z - w| + |z|.$$

The **complex conjugate** of $z = x + iy$ is defined by

$$\overline{z} = x - iy,$$

and it is obtained by a reflection across the real axis in the plane. In fact a complex number $z$ is real if and only if $z = \overline{z}$, and it is purely imaginary if and only if $z = -\overline{z}$.

The reader should have no difficulty checking that

$$\mathrm{Re}(z) = \frac{z+\overline{z}}{2} \quad \text{and} \quad \mathrm{Im}(z) = \frac{z-\overline{z}}{2i}.$$

Also, one has

$$|z|^2 = z\overline{z} \quad \text{and as a consequence} \quad \frac{1}{z} = \frac{\overline{z}}{|z|^2} \quad \text{whenever } z \neq 0.$$

Any non-zero complex number $z$ can be written in **polar form**

$$z = re^{i\theta},$$

where $r > 0$; also $\theta \in \mathbb{R}$ is called the **argument** of $z$ (defined uniquely up to a multiple of $2\pi$) and is often denoted by $\arg z$, and

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Since $|e^{i\theta}| = 1$ we observe that $r = |z|$, and $\theta$ is simply the angle (with positive counterclockwise orientation) between the positive real axis and the half-line starting at the origin and passing through $z$. (See Figure 2.)

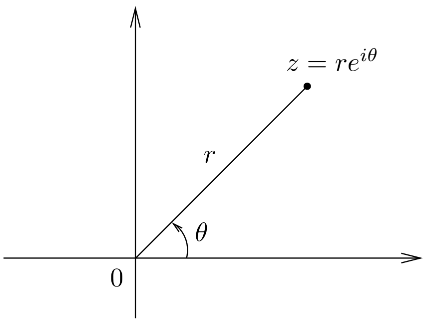


**Figure 2.** The polar form of a complex number

Finally, note that if $z = re^{i\theta}$ and $w = se^{i\varphi}$, then

$$zw = rse^{i(\theta+\varphi)},$$

so multiplication by a complex number corresponds to a homothety in $\mathbb{R}^2$ (that is, a rotation composed with a dilation).

## 1.2 Convergence

We make a transition from the arithmetic and geometric properties of complex numbers described above to the key notions of convergence and limits.

A sequence $\{z_1, z_2, \ldots\}$ of complex numbers is said to **converge** to $w \in \mathbb{C}$ if

$$\lim_{n\to\infty} |z_n - w| = 0, \quad \text{and we write} \quad w = \lim_{n\to\infty} z_n.$$

This notion of convergence is not new. Indeed, since absolute values in $\mathbb{C}$ and Euclidean distances in $\mathbb{R}^2$ coincide, we see that $z_n$ converges to $w$ if and only if the corresponding sequence of points in the complex plane converges to the point that corresponds to $w$.

As an exercise, the reader can check that the sequence $\{z_n\}$ converges to $w$ if and only if the sequence of real and imaginary parts of $z_n$ converge to the real and imaginary parts of $w$, respectively.

Since it is sometimes not possible to readily identify the limit of a sequence (for example, $\lim_{N\to\infty}\sum_{n=1}^{N} 1/n^3$), it is convenient to have a condition on the sequence itself which is equivalent to its convergence. A sequence $\{z_n\}$ is said to be a **Cauchy sequence** (or simply **Cauchy**) if

$$|z_n - z_m| \to 0 \quad \text{as } n, m \to \infty.$$

In other words, given $\epsilon > 0$ there exists an integer $N > 0$ so that $|z_n - z_m| < \epsilon$ whenever $n, m > N$. An important fact of real analysis is that $\mathbb{R}$ is complete: every Cauchy sequence of real numbers converges to a real number.[1] Since the sequence $\{z_n\}$ is Cauchy if and only if the sequences of real and imaginary parts of $z_n$ are, we conclude that every Cauchy sequence in $\mathbb{C}$ has a limit in $\mathbb{C}$. We have thus the following result.

**Theorem 1.1** $\mathbb{C}$, *the complex numbers, is complete.*

We now turn our attention to some simple topological considerations that are necessary in our study of functions. Here again, the reader will note that no new notions are introduced, but rather previous notions are now presented in terms of a new vocabulary.

## 1.3 Sets in the complex plane

If $z_0 \in \mathbb{C}$ and $r > 0$, we define the **open disc** $D_r(z_0)$ **of radius** $r$ **centered at** $z_0$ to be the set of all complex numbers that are at absolute

[1] This is sometimes called the Bolzano-Weierstrass theorem.